Parking Space Availability Prediction Using Random Forest and Time-Series Models

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

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TEAM 17

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (CSE)

Name of Title: Parking Space Availability Prediction Using Random Forest and Time-Series Models

1. Abstract

Parking space availability is a major problem in crowded urban areas, where drivers often spend considerable time searching for empty parking spaces. This results in traffic congestion, fuel wastage, air pollution, and unnecessary travel delays. This project, **Parking Space Availability Prediction Using Random Forest and Time-Series Models**, aims to develop a machine learning-based system that predicts parking availability for a given location and time.

The project uses historical parking data containing information such as parking location, parking capacity, occupancy, and date and time of observation. The dataset is preprocessed by handling missing values, extracting useful time-based features, and preparing the data for machine learning. The Random Forest algorithm is used as the primary supervised learning model to predict parking availability based on historical parking patterns and relevant features. Random Forest is selected because it performs well on structured data and can handle multiple features effectively.

In addition to Random Forest, Time-Series analysis is used to study changes in parking occupancy over time and forecast future parking demand. Features such as hour, day, weekday, and occupancy patterns help the system identify peak and low-demand periods. The predicted results can be presented through a simple web application or dashboard,

allowing users to check parking availability before reaching their destination.

The proposed system can help drivers save time and fuel while reducing unnecessary traffic caused by searching for parking. It can also help parking management authorities understand demand patterns and utilize available parking spaces more efficiently. In the future, the system can be enhanced using real-time sensor data, live parking information, GPS, and mobile applications. Thus, the project provides a practical machine learning solution for smarter and more efficient parking management.

2. INTRODUCTION

The increasing number of vehicles in cities has created a growing demand for efficient parking management. Finding an empty parking space can be difficult, especially during peak hours. Drivers may spend several minutes searching for parking, which increases traffic congestion, fuel consumption, and pollution.

The proposed project uses machine learning to predict parking space availability based on historical parking data. The system uses **Random Forest** to predict parking availability and **Time-Series models** to understand and forecast parking occupancy patterns over time. The input data includes parking capacity, occupancy, location, date, and time.

The system follows a simple process: first, the parking dataset is collected and cleaned. Next, useful features are extracted from the date and time information. The processed data is given to the machine learning models for training and prediction. Finally, the predicted parking status can be displayed through a web-based dashboard.

This project can benefit both drivers and parking authorities. Drivers can use the prediction to make better decisions about where and when to park, while authorities can use the information to understand parking demand and improve parking management.

3. SOFTWARE REQUIREMENTS

Software Requirements

- **Operating System:** Windows / Linux
- **Programming Language:** Python
- **Development Environment:** Jupyter Notebook / Google Colab / VS Code
- **Libraries:** Pandas, NumPy
- **Machine Learning:** Random Forest
- **Time-Series Analysis:** Python Time-Series libraries
- **Dataset:** Parking Birmingham dataset
- **Browser:** Google Chrome / Microsoft Edge

Signature of the Guide **Course Instructor**